

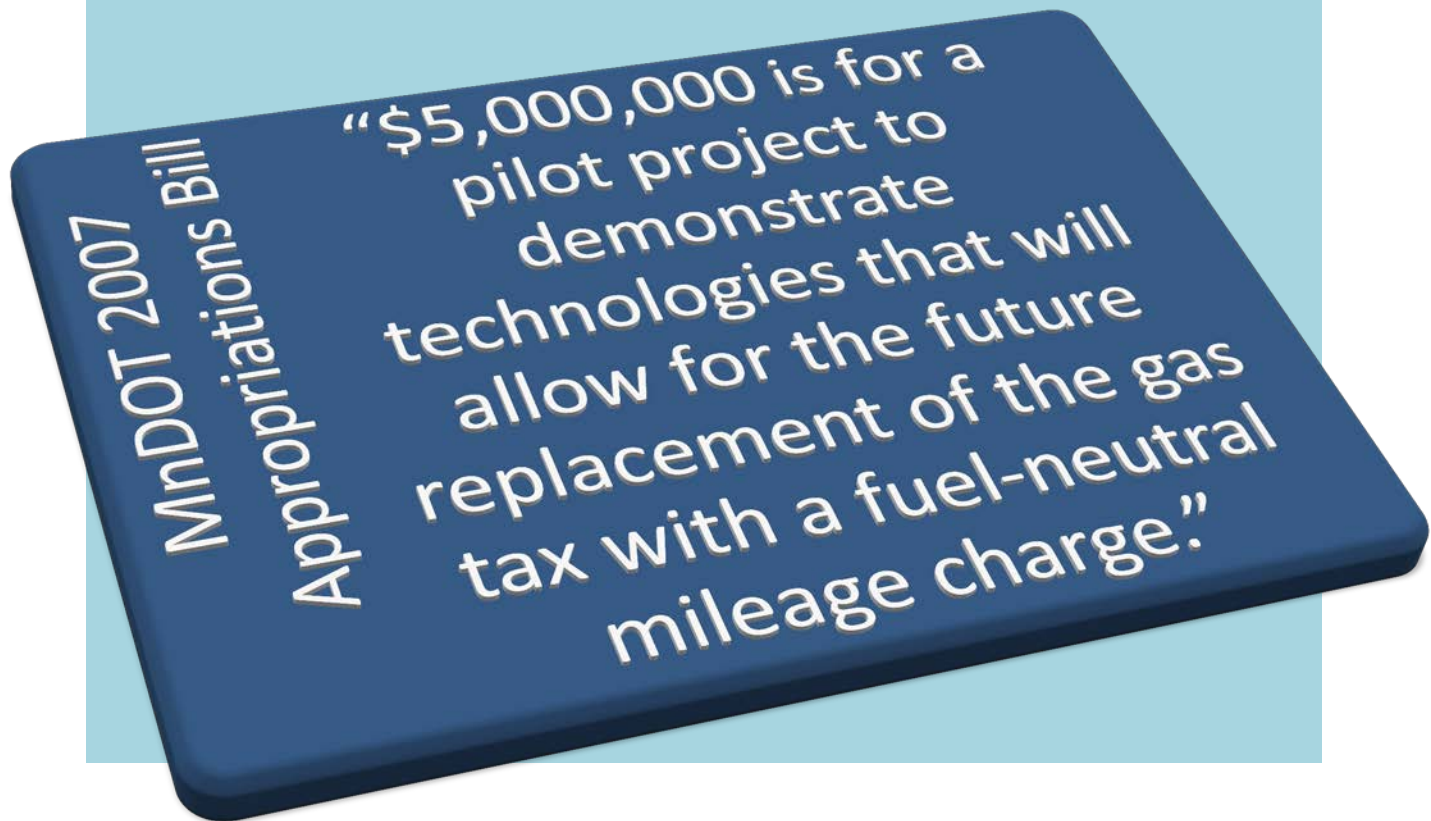


“MBUF Demo” “Mn Road Fee Test” “IntelliDrive Connected Vehicles for Safety, Mobility and User Fee”

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Office of Traffic, Safety, and Technology
MnDOT



Statutory Direction



State and Federal Gas Taxes Paid Annually

(taxes per year)	Light Duty Truck (20 mpg)		Passenger Car (30 mpg)		Hybrid (40 mpg)		Electric Vehicle (non-gas powered)	
	State Tax *	Federal Tax **	State Tax *	Federal Tax **	State Tax *	Federal Tax **	State Tax *	Federal Tax **
20,000 miles/year	\$280	\$184	\$187	\$123	\$140	\$92	\$0	\$0
15,000 miles/year	\$210	\$138	\$140	\$92	\$105	\$69	\$0	\$0
10,000 miles/year	\$140	\$92	\$93	\$61	\$70	\$46	\$0	\$0
Equivalent cents/mile	1.40	0.92	0.93	0.61	0.70	0.46	0	0

* Minnesota tax on gasoline is \$0.28 per gallon as of publication.

** Federal tax on gasoline is \$0.184 per gallon as of publication.

Miles Based User Fees: a Two-Part Research Effort

1. Technology Demonstration

- 500 volunteers from Wright County, MN
- Existing smart phone and cellular communications

2. Policy Study

- Engage key stakeholders to identify and evaluate MBUF issues
- Report policy level recommendations to MnDOT Commissioner



Demonstration Approach

- Battelle to design, build, and test the technology
- SAIC to recruit the volunteers and evaluate the demonstration
- Three test waves of approximately 175 each

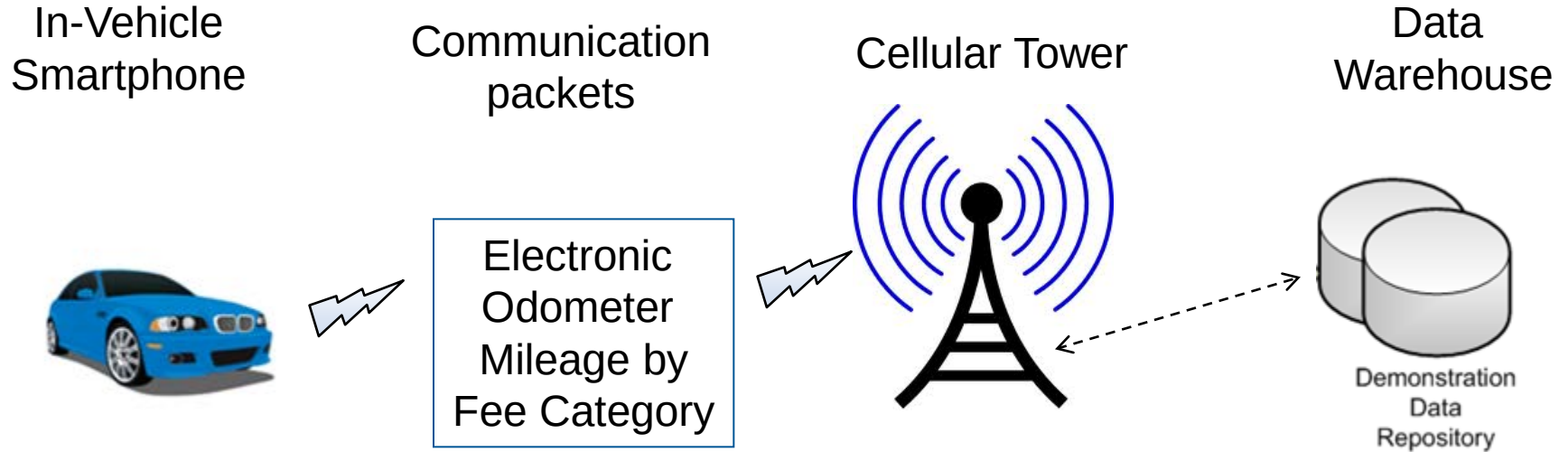
WAVE	Fall		Winter					Spring		Summer			Fall	
	SEPT	OCT	NOV	DEC	JAN	FEB	MAR	APR	MAY	JUN	JUL	AUG	SEPT	OCT
A	BASELINE		TEST PERIOD											
B					BASELINE		TEST PERIOD							
C									BASELINE		TEST PERIOD			

- Final report scheduled to be released - Dec. 2012

Technology Requirements

- Develop software and hardware systems that:
 - Using Commercial Off-The-Shelf (COTS) hardware
 - Deploying 500 in-vehicle devices & back office processing system
 - Capturing, invoicing, and handling payments associated with MBUFs
 - Permit a comparison of technology vs. odometer readings
 - Also deploy Value Added Services
 - Navigation, Safety Signage & “Trip” Data Collection

System Components



- GPS Position is NOT being transmitted for the Fee Assessment
- Website access to data warehouse elements

In-Vehicle Hardware

Samsung Galaxy S

In-Vehicle
Computational
Device and
Display Unit



OBD-II Blue tooth Tool

STN1110
With Bluetooth
Data Connection



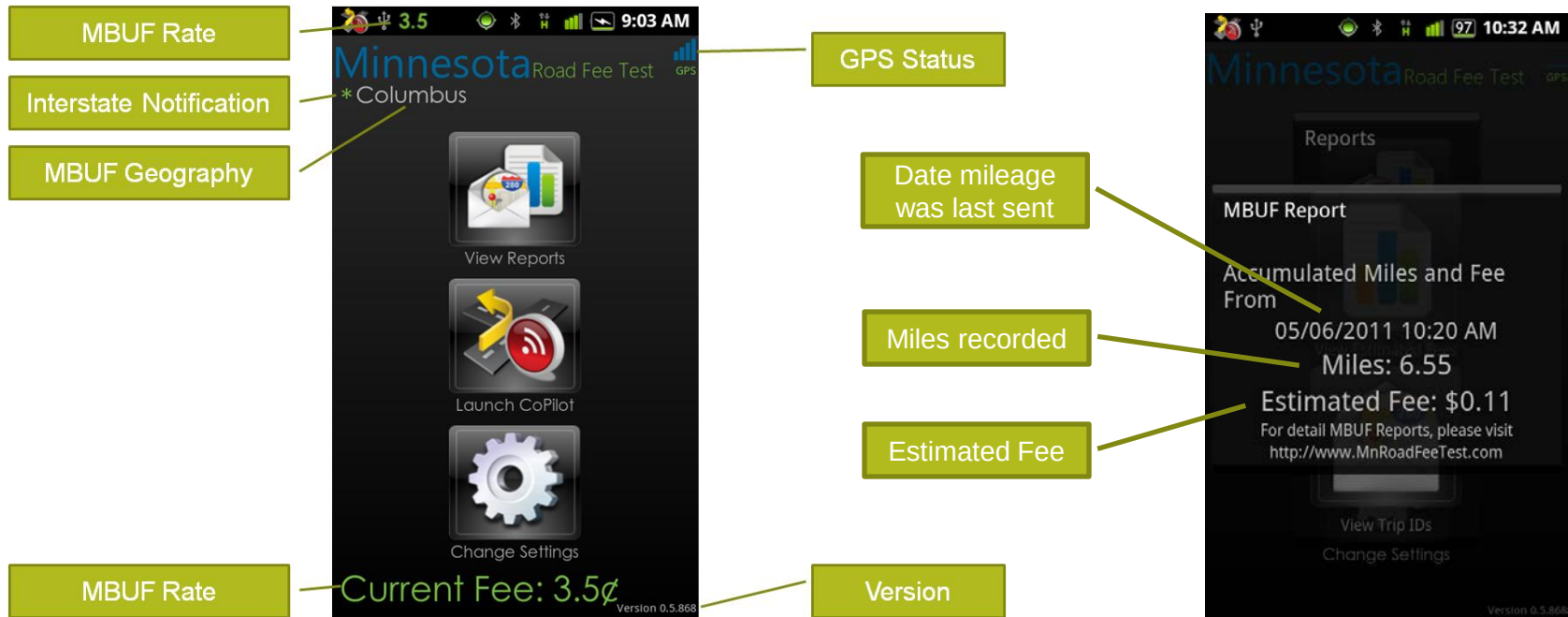
COTS Mounting Hardware and Cables



MBUF – How it works

- Odometer reading before and after testing
- Technology is voluntarily used to discount rate
- MBUF application accumulates mileage driven by fee category (roadway type, geography, etc.)
- Every 24 hours accumulated mileage by fee category transmitted via cellular to data warehouse
- Invoice generated monthly based upon aggregate miles in fee categories
- Participant pays monthly billing

What Does a Participant See?



Electronic and Paper Invoices

Minnesota Road Fee Test

Monthly Invoice

Matthew G Burns
505 King Ave
Columbus, OH 43201

Invoice #: 2099
2002 VW Passat

Invoice Date: 6/9/2011

Due Date: 6/22/2011

Total Amount Due: \$0.46

Mileage Detail

Criteria	Mileage	Rate (\$)	Rate Effectivity Date	Total Charge
Columbus-NonInterstate-OFF-WEEKDAY	20.23	0.010	6/8/2011	\$0.20
Columbus-NonInterstate-AM-WEEKDAY	6.95	0.020	6/8/2011	\$0.14
Ohio-NonInterstate-OFF-WEEKDAY	5.46	0.010	4/20/2011	\$0.05
Ohio-NonInterstate-AM-WEEKDAY	3.70	0.010	4/22/2011	\$0.04
Bartholomew North-NonInterstate-AM-WEEKDAY	0.19	0.080	6/5/2011	\$0.02
Bartholomew South-NonInterstate-OFF-WEEKDAY	0.20	0.070	6/5/2011	\$0.01
Columbus-Interstate-AM-WEEKDAY	0.04	0.035	4/22/2011	\$0.00
Columbus-Interstate-OFF-WEEKDAY	0.03	0.035	4/22/2011	\$0.00
Columbus-NonInterstate-PM-WEEKDAY	0.01	0.020	6/8/2011	\$0.00
Battelle Parking Lot-NonInterstate-AM-WEEKDAY	0.03	0.050	5/9/2011	\$0.00
Battelle Parking Lot-NonInterstate-OFF-WEEKDAY	0.09	0.050	5/9/2011	\$0.00

Payment Detail

Last Payment Amount	Last Payment Date
\$0.47	5/24/2011

To make a payment using PayPal, logon to <https://www.mnroadfeetest.com>.

Payments also may be made by appointment by calling (612) 708-7288.

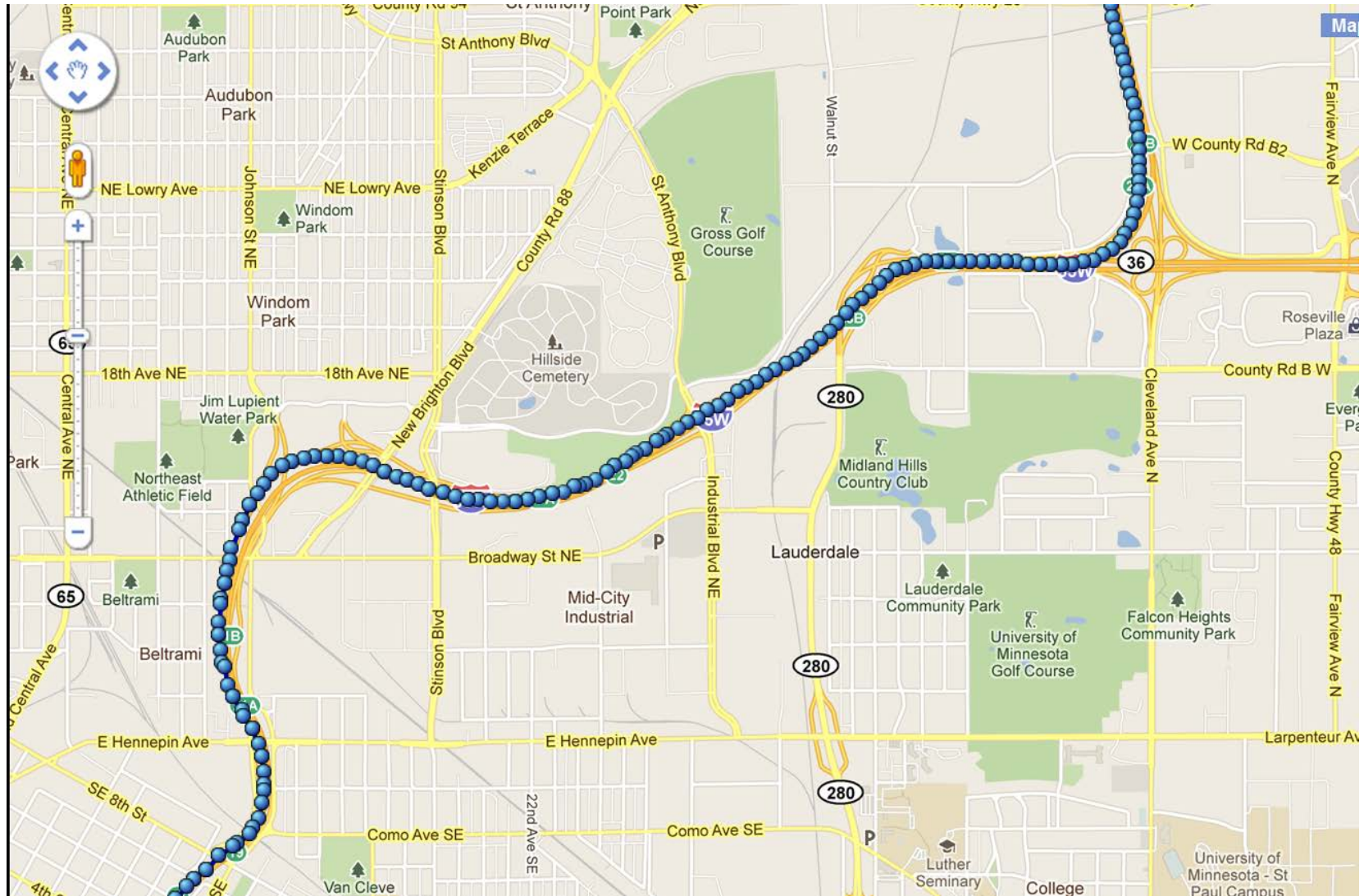
For questions about this invoice, call (612) 708-7288 or email billing@mnroadfeetest.com

Disputing the Fee Calculation Option

- “Trip” application captures vehicle location every second and sends this to infrastructure every 20 seconds.
- “Trip” does not contain ability to link a participant to a route, **only a participant can do this**

MBUF Application	Trip Application
• Stores accumulated miles by road rate category in OBU • Transmits cumulative miles by category and vehicle ID to Infrastructure no more than once per day • No information on individual trips	• Latitude/Longitude on second-by-second basis • Transmitted to Infrastructure every 20 seconds • Contains a TRIP identifier (ID) but no information on vehicle or person

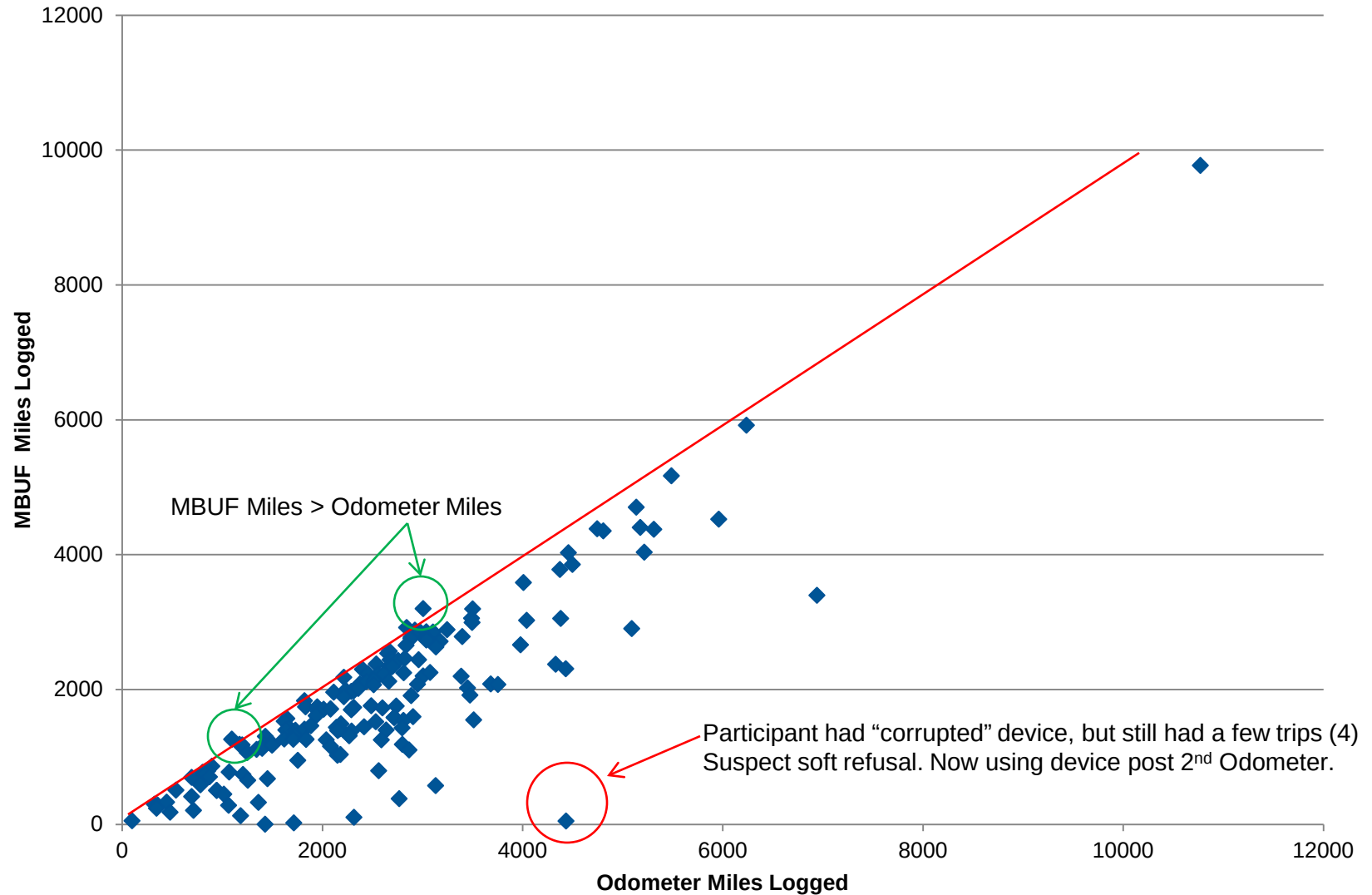
Reported Individual Trip Info



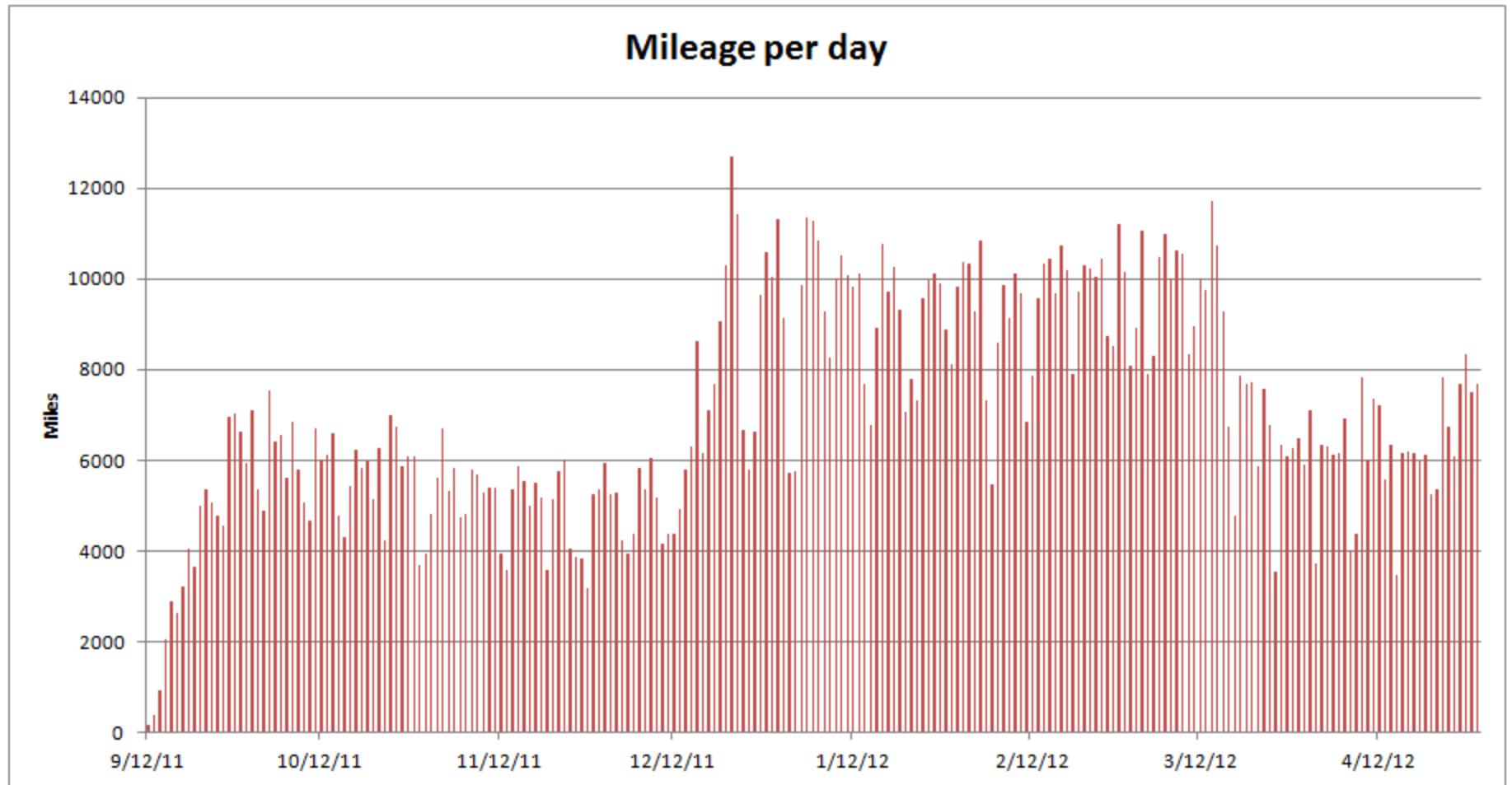
Status of Evaluation Activities

Activities	Wave A	Wave B	Wave C
Recruitment	196 recruited 146 began test in Sept 137 completed the test <i>2 participants pending</i>	210 recruited 165 began test in Dec 162 currently active	260 Recruited 189 Scheduled for Odometer Reading #1
Survey #1	Complete 142 out of 146 (97%)	Complete 162 out of 165 (98%)	Live: 5/14-5/28
Survey #2	Complete 136 out of 141 (96%)	Complete 157 out of 162 (97%)	Live: 7/11-7/25
Survey #3	Complete 132 out of 139 (95%)	Live: 5/21-6/4	Live: 10/3-10/17
Focus Groups	Completed 2/28 & 2/29	May 22 and 23	early Oct
Observing Odometer Readings	Observed #1 9/20-9/21 Observed #2 11/21-11/22 Observed #3 3/20-3/21	-	-
One-on-one Interviews	Complete 127 out of 139 (91%)	May / June	Oct / Nov

Odometer versus MBUF: Wave B



Mileage Collection



MRFT Mileage Invoicing

- As of the end of March we have invoiced:
 - 1,418,166 miles across 2184 invoices worth \$11,996.31
- 658,830 Baseline miles (\$0.00 / mile)
- 759,336 Test miles (\$0.00, \$0.01 or \$0.03 / mile)

How do I know what fees I will be charged if I'm using the device?

Ask three questions...

1. Where am I currently driving?
2. What day of the week am I driving?
3. At what time of day am I driving?

Current Driving Location	Peak Times Monday, Tuesday, Wednesday, Thursday, Friday 7AM-9AM 4PM-6PM	Off Peak Times
Outside of Minnesota	\$.00	\$.00
Inside Minnesota and Outside the Twin Cities Metro Zone	\$.01	\$.01
Inside the Twin Cities Metro Zone	\$.03	\$.01

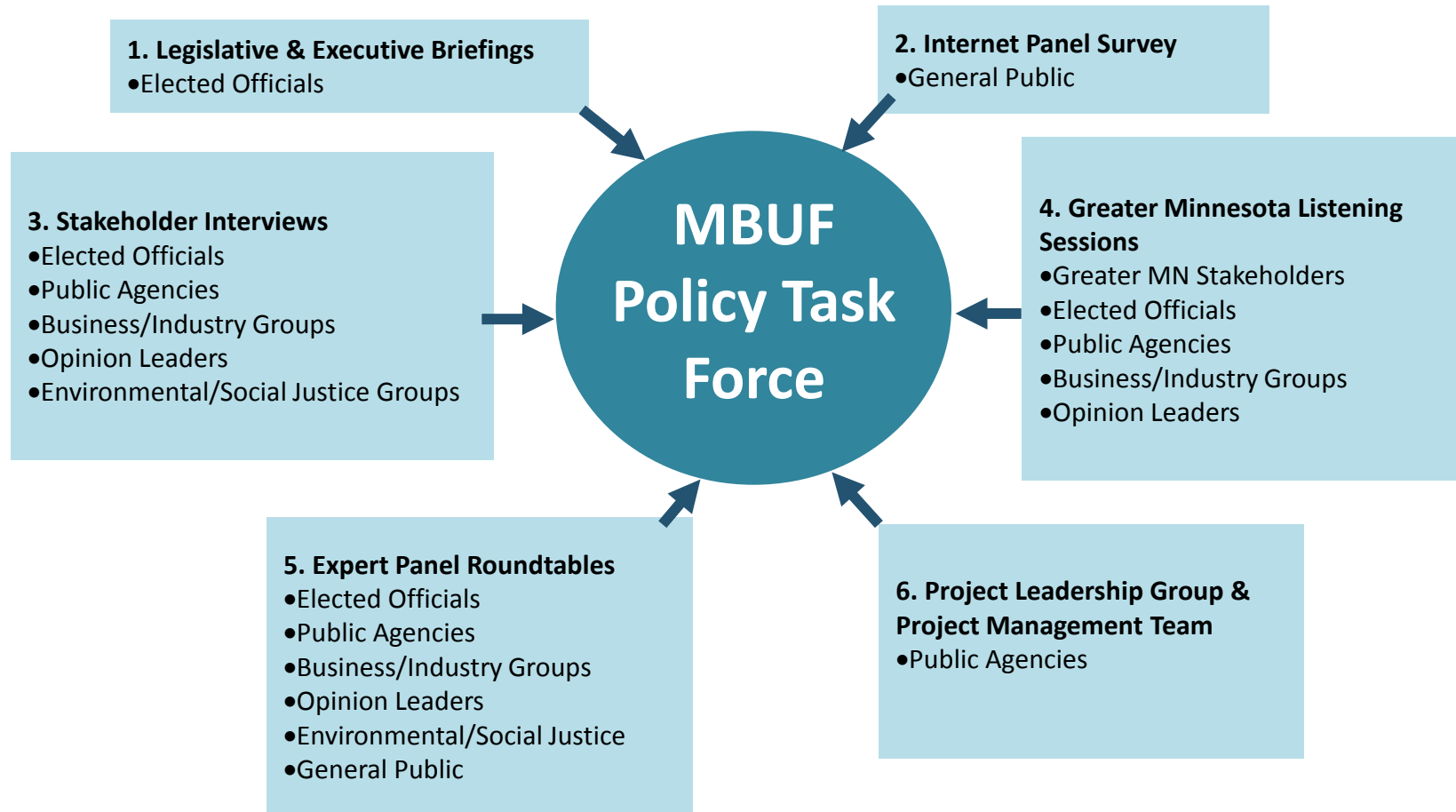
Preliminary Operational Conclusions

- GPS technology via smart phone is a potentially viable technology mechanism for MBUF provided:
 - Mechanism to also capture odometer
 - Ability to handle cases when GPS not available
- Smart Phone platform is relatively stable and is able to meet the processing demands
- Monthly billing seems to be acceptable to participants
 - A few participants have expressed less frequent billing would be better
- Ability to see trip details in relation to fees assessed is of great interest by participants

MBUF Policy Study

- Hired the University of MN to identify and evaluate issues of an MBUF system
- 25 person task force was assembled and held monthly meetings during the summer and fall of 2011
- Task force prepared and presented a report to the Commissioner of Transportation

Stakeholder Input Approach



MBUF Task Force Members



Front row: Margaret Donahoe, Mayor Jim Hovland (Vice Chair), Bernie Lieder (Chair), Steve Elkins, Ron Erhardt, Wade Kline.

Second row: State Representative Frank Hornstein, State Senator Scott Dibble, Steve Murphy, Bob Anderson, Christine Zimmer.

Third row: Barb Thoman, Roy Terwilliger, Bob DeBoer, State Representative Mike Beard, John Mock

Back row: Bob Krogman, John Hausladen, Donald Jensen.

Task Force Results

Report presented to MnDOT Commissioner on December 5, 2011

- Part 1: Describes what the Task Force learned
 - Benefits and concerns
 - Problem statement and objectives
- Part 2: Offers the Task Force's findings and recommendations
 - Six findings and recommendations

Problem Statement

“As more people continue to use fuel-efficient and alternative-fuel vehicles that are not fully taxed or are untaxed, less revenue will be generated by the fuel tax.

In addition, changes in demographics and travel trends will further reduce revenue contributed to the fuel tax fund.

As a result, future revenues will be inadequate to fund Minnesota’s transportation infrastructure.”

Policy Objectives

PRIMARY OBJECTIVES

- Promote Equity
- Generate Transportation Funds

ANCILLARY LONG TERM OBJECTIVES

- Protect the Environment
- Improve Transportation System Performance

Recommendations

Recommendation #1: All drivers pay their fair share

Recommendation #2: Include all vehicles using roadway system

Recommendation #3: State policy makers continue to discuss MBUF

Recommendation #4: State government continue to conduct trials

Recommendation #5: MnDOT conduct further technical analysis

Recommendation #6: MnDOT provide policy makers with results and how to setup future MBUF systems

Policy Conclusion

Task Force recommends that exploration of an MBUF system for Minnesota continue to advance in a measured, informed and thoughtful manner guided by the six findings and recommendations.

Report, including a minority opinion, was unanimously adopted by Task Force



Minnesota's MBUF: Summary

1. Technology Demonstration

- Field test is going well and about 80% complete
- Technology is working very well and should deliver a wealth of detailed data
- Final Report expected Dec 2012

2. Policy Study

- Legislative and key stakeholders have been engaged
- Task force report completed
- Transportation finance advisory committee effort started





Minnesota Mileage Based User Fee Program

QUESTIONS or COMMENTS?

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Reports Available at:

<http://www.dot.state.mn.us/mileagebaseduserfee/index.html>

